Supporting Information

1 Observational atmospheric temperature data

We used satellite measurements of atmospheric temperature change produced by RSS (S1) and UAH (S2). Both groups provide MSU-based estimates of vertically weighted, layer-average temperature changes for TLS, TMT, and TLT. We analyzed MSU temperatures from data set versions 3.3 (RSS) and 5.5 (UAH). Observed MSU data were in the form of monthly means for the 408-month period January 1979 to December 2012, and are on a $2.5^\circ \times 2.5^\circ$ latitude/longitude grid.

2 Model simulation output

We analyzed atmospheric temperature changes from the following types of simulation: ALL+8.5 (20 models), CTL (20 models), NAT (16 models), P1000 (6 models), ANT (8 models), SOL (3 models), and VOL (2 models). Tables S1-S9 of the *SI Appendix* provide information on the models used in our analysis, their horizontal and vertical resolution, applied external forcings, splicing, and simulation start and end dates.

2.1 Splicing information

For 19 of the 20 CMIP-5 models analyzed here, the ALL synthetic MSU temperature changes were spliced together with RCP8.5 results. In the case of `CNRM-CM5`, there is a discontinuity in the volcanic aerosol forcing between the ALL and RCP8.5 simulations. This discontinuity was removed by splicing the `CNRM-CM5` ALL and historicalExt simulations (S3).

2.2 Treatment of different GISS-E2-R physics versions

For the `GISS-E2-R` ALL+8.5 simulations, output was available from two different model versions (`p1` and `p2`). In calculating ALL+8.5 multi-model averages, we treated atmospheric temperatures from the `p1` and `p2` ALL+8.5 runs as results from two different models of the climate system rather than as different realizations of climate change performed with a similar physical model (S3).

3 RSS percentile realizations

Our S/N analysis relied on both the publicly-available version of the RSS atmospheric temperature data sets and on RSS estimates of observational uncertainty. Uncertainty estimates account for "sampling error, premerge adjustments to each individual satellite, and the merging procedure", and were obtained using a Monte Carlo approach (S1). For each atmospheric layer considered here (TLS, TMT, and TLT), a 400-member ensemble of gridded, monthly-mean temperature data sets was generated. To reduce the ensemble size, we ranked the 400 realizations by their global-mean temperature trends over 1979 to 2012, and then chose 11 realizations for each atmospheric layer. Ranking was performed separately for TLS, TMT, and TLT. The selected realizations were the 5^{th}, 10^{th}, 20^{th}, 30^{th}, 40^{th}, 50^{th}, 60^{th}, 70^{th}, 80^{th}, 90^{th}, and 95^{th} percentiles of the ranked distributions.

4 Regridding of model and observational data

For the S/N analysis, it was necessary to transform temperature data from the observational grid and the original grids of the CMIP-5 models to a common grid. Prior to calculation of zonal averages, we regridded all data sets to a $5^\circ \times 5^\circ$ latitude/longitude grid. Model and UAH temperatures were masked using the RSS spatial coverage, and then extracted for 80°N-80°S (TLS and TMT), and 80°N-70°S (TLT).

5 Fingerprint estimation

Let $S(i, j, x, h, t)$ represent annual-mean synthetic MSU temperature data at latitude band x, layer h, and time t from the i^{th} realization of the j^{th} model's ALL+8.5 or ANT simulation, where:

$i = 1, \ldots N_r(j)$ (the number of realizations for the j^{th} model).

$j = 1, \ldots N_m$ (the number of models used in fingerprint estimation).

$x = 1, \ldots N_x$ (the total number of latitude bands).

$h = 1, \ldots N_h$ (the total number of atmospheric layers).

$t = 1, \ldots N_t$ (the time in years).

Here, $N_r(j)$ is 1 for the ANT runs, and ranges from 1 to 5 for the ALL+8.5 simulations; $N_m = 20$ (ALL+8.5), or 8 (ANT); $N_x = 33$ latitudes; N_h is generally 3 layers; and N_t is either 152 years (ALL+8.5) or 145 years (ANT).

As in previous work (S3, S4), we define the fingerprint $F(x,h)$ as the first EOF of the covariance matrix of $\overline{\overline{S}}(x,h,t)$. Note that $\overline{\overline{S}}(x,h,t)$ was calculated by first averaging over an individual model's realizations (where multiple realizations were available), and then averaging over models. Because atmospheric temperature changes in most CMIP-5 CTL simulations do not exhibit pronounced residual drift (S3), CTL information was not subtracted from the ALL+8.5 simulations prior to fingerprint calculations.

Many fingerprint detection strategies seek to rotate $F(x,h)$ in a direction that maximizes the signal strength relative to the control run noise (S5, S6, S7). Optimization of $F(x,h)$ generally leads to enhanced detectability of the signal. In almost all cases we considered, we achieved detection of an externally-forced fingerprint in observations without any optimization of $F(x,h)$. We therefore report only on S/N ratios obtained with non-optimized fingerprints.

To explore the sensitivity of our S/N results to different plausible choices in the fingerprint estimation process, we computed $F(x,h)$ using input atmospheric temperature data over 1861 to 2012 and 1979 to 2012 for the ALL+8.5 simulations and over 1861 to 2005 and 1979 to 2005 for the shorter ANT runs. Positive identification of the ALL+8.5 and ANT fingerprints in observations is insensitive to these choices.

6 Calculation of concatenated noise data sets

We obtain V_{INT} estimates from concatenated CTL runs; V_{TOT} is estimated using either concatenated NAT simulations or concatenated P1000 integrations. Because the length of the 20 CTL runs analyzed here varies by a factor of up to 4, models with longer control integrations could have a disproportionately large impact on the V_{INT} estimates. To guard against this possibility, our V_{INT} estimates rely on only the last 200 years of each model's pre-industrial control run. The total number of years of concatenated data is therefore 4000 in the CTL case. Use of the full (unequal length) CTL simulations does not yield noticeably different S/N results.

Each of the 16 NAT integrations spans a similar period (see Table S5), so it is not necessary to extract NAT segments of equal length. The total number of concatenated NAT years is 2549. For the P1000 simulations, we used years 850 to 1700 for each of 6 models, yielding 5106 years of concatenated data. Use of the full (850 to 1849) P1000 information slightly increases V_{TOT} levels, but does not affect anthropogenic fingerprint detection.

Annual-mean synthetic MSU temperatures from individual model CTL, NAT, and P1000 runs were regridded to the $5^\circ \times 5^\circ$ target grid used for fingerprint estimation. After regridding, anomalies were defined relative to climatological annual means over: 1) the full length of each CTL run; 2) 1861 to 2005 (NAT); and 3) 850 to 1700 (P1000). Anomalies were then zonally averaged.

Two of the 20 CTL runs analyzed here have some small residual drift (S3). We assume that the drift behavior of synthetic MSU temperature can be well-approximated by a least-squares linear trend, and remove drift at each latitude band. Drift removal is performed over the last 200 CTL years (since only the last 200 years are concatenated). Because it is difficult to discriminate between residual drift and low-frequency changes in externally forced natural variability, no drift is removed from the NAT or P1000 simulations. Any residual drift inflates total natural variability, leading to conservative estimates of S/N ratios.

7 Estimating S/N ratios and detection time

We begin with $O(x,h,t)$, the regridded annual-mean, zonal-mean TLS, TMT, and TLT data from UAH, RSS, and the 11 RSS percentile realizations. Observed data are expressed as anomalies relative to climatological annual means over 1979 to 2012. The observations are then projected onto $F(x,h)$, the time-invariant fingerprint:

$$c\{F,O\}(t) = \sum_{x=1}^{N_x} \sum_{h=1}^{N_h} F(x,h)\, O(x,h,t) \tag{1}$$

This projection is equivalent to a spatially uncentered covariance, $c\{F,O\}(t)$, between the patterns $F(x,h)$ and $O(x,h,t)$ at time t. The signal time series $c\{F,O\}(t)$ provides information on the amplitude of the fingerprint in the observational data.

To assess trend significance, we require a case in which $O(x,h,t)$ is replaced by model-based estimates of either V_{INT} or V_{TOT}. Here, we use the CTL, NAT, and

P1000 natural variability data sets, $N(x,h,t)$, which have been regridded, detrended, and concatenated as described above. The time series $c\{F,N\}(t)$ is the spatially uncentered covariance of $F(x,h)$ and $N(x,h,t)$:

$$c\{F,N\}(t) = \sum_{x=1}^{N_x} \sum_{h=1}^{N_h} F(x,h)\, N(x,h,t) \tag{2}$$

We estimate S/N ratios by fitting least-squares linear trends of increasing length L years to $c\{F,O\}(t)$, and then comparing these with $s(L)$, the standard deviation of the distribution of non-overlapping L-length trends in $c\{F,N\}(t)$. Because of the episodic nature of volcanially-induced temperature signals, use of maximally overlapping trends (*i.e.*, with overlap by all but one year) slightly inflates estimates of $s(L)$ in the NAT and P1000 cases, but does not negate detection of the ANT or ALL+8.5 fingerprints.

Signal detection is stipulated to occur when the trend in $c\{F,O\}(t)$ exceeds and remains above some chosen significance level (typically 1%). The test is one-tailed, and we assume a Gaussian distribution of trends in $c\{F,N\}(t)$. The start date for fitting linear trends to $c\{F,O\}(t)$ is 1979, the first complete year of the observational MSU data. We use a minimum trend length of ten years, so the first S/N ratio is for 10-year trends ending in 1988. The last S/N ratio is for 34-year trends ending in 2012.

References for SI Appendix

S1. Mears C, Wentz FJ, Thorne P, Bernie D (2011) Assessing uncertainty in estimates of atmospheric temperature changes from MSU and AMSU using a Monte-Carlo technique. *J Geophys Res* 116, D08112, doi:10.1029/2010JD014954.

S2. Christy JR, Norris WB, Spencer RW, Hnilo JJ (2007) Tropospheric temperature change since 1979 from tropical radiosonde and satellite measurements. *J Geophys Res* 112, D06102, doi:10.1029/2005JD006881.

S3. Santer BD, *et al.* (2013) Identifying human influences on atmospheric temperature. *Proc Nat Acad Sci* 110:26-33.

S4. Santer BD, *et al.* (2003) Contributions of anthropogenic and natural forcing to recent tropopause height changes. *Science* 301:479-483.

S5. Tett SFB, *et al.* (2002) Estimation of natural and anthropogenic contributions to twentieth century temperature change. *J Geophys Res* 107, D16, doi:10.1029/2000JD000028.

S6. Gillett NP, *et al.* (2011) Attribution of observed changes in stratospheric ozone and temperature. *Atmos Chem Phys* 11:599-609.

S7. Hasselmann K (1979) On the signal-to-noise problem in atmospheric response studies. In *Meteorology of Tropical Oceans*, Shaw DB (ed), pp. 251-259, Roy Met Soc London, U.K.

S8. Eyring V, *et al.* (2013) Long-term changes in tropospheric and stratospheric ozone and associated climate impacts in CMIP5 simulations. *J Geophys Res* (in press).

S9. Schmidt GA, *et al.* (2012) Climate forcing reconstructions for use in PMIP simulations of the Last Millennium. *Geosci Model Dev* 5:185-191.

Table S1: Modeling center information and official acronyms of the CMIP-5 models used in this study.

	Model	**Country**	**Modeling center**
1	BCC-CSM1.1	China	Beijing Climate Center, China Meteorological Administration
2	CanESM2	Canada	Canadian Centre for Climate Modelling and Analysis
	Ensemble member ID: p1, p3		
3	CCSM4	USA	National Center for Atmospheric Research
4	CNRM-CM5	France	Centre National de Recherches Meteorologiques / Centre Européen de Recherche et Formation Avancées en Calcul Scientifique
5	CSIRO-Mk3.6.0	Australia	Commonwealth Scientific and Industrial Research Organization in collaboration with Queensland Climate Change Centre of Excellence
	Ensemble member ID: p1, p6		
6	FGOALS-g2	China	LASG, Institute of Atmospheric Physics, Chinese Academy of Sciences; and CESS, Tsinghua University
7	GFDL-CM3	USA	NOAA Geophysical Fluid Dynamics Laboratory
	Ensemble member ID: p1, p2		
8	GFDL-ESM2G	USA	NOAA Geophysical Fluid Dynamics Laboratory
9	GFDL-ESM2M	USA	NOAA Geophysical Fluid Dynamics Laboratory
	Ensemble member ID: p1, p2, p7, p8		
10	GISS-E2-H	USA	NASA Goddard Institute for Space Studies
	Ensemble member ID: p1, p109		
11	GISS-E2-R	USA	NASA Goddard Institute for Space Studies
	Ensemble member ID: p1, p2, p3, p102, p109, p121, p309		
12	HadGEM2-CC	UK	Met. Office Hadley Centre

Table S1: Modeling center information and official acronyms of the CMIP-5 models used in this study (continued).

	Model	**Country**	**Modeling center**
13	HadGEM2-ES	UK	Met. Office Hadley Centre
14	INM-CM4	Russia	Institute for Numerical Mathematics
15	IPSL-CM5A-LR Ensemble member ID: p1, p2	France	Institut Pierre-Simon Laplace
16	MIROC5	Japan	Atmosphere and Ocean Research Institute (the University of Tokyo), National Institute for Environmental Studies, and Japan Agency for Marine-Earth Science and Technology
17	MIROC-ESM-CHEM	Japan	Japan Agency for Marine-Earth Science and Technology, Atmosphere and Ocean Research Institute (the University of Tokyo), and National Institute for Environmental Studies
18	MIROC-ESM	Japan	Japan Agency for Marine-Earth Science and Technology, Atmosphere and Ocean Research Institute (the University of Tokyo), and National Institute for Environmental Studies
19	MPI-ESM-LR	Germany	Max Planck Institute for Meteorology
20	MPI-ESM-P	Germany	Max Planck Institute for Meteorology
21	MRI-CGCM3	Japan	Meteorological Research Institute
22	NorESM1-M	Norway	Norwegian Climate Centre

Table S2: Information on horizonal resolution (latitude/longitude), the number of atmospheric levels ($N_{\rm H1}$)[1], the top model level (in hPa), the number of atmospheric levels above 100 hPa ($N_{\rm H2}$)[2], and the availability of simulation output for the CMIP-5 models used in this study (A = ALL+8.5; B = CTL; C = NAT; D = P1000; E = ANT; F = VOL; G = SOL).

	Model	Resolution	$N_{\rm H1}$	Top	$N_{\rm H2}$	A	B	C	D	E	F	G
1	BCC-CSM1.1	64 × 128	17	10	5	X	X	X	X	-	-	-
2	CanESM2 (p1)	64 × 128	22	1	10	X	X	X	-	-	-	-
3	CanESM2 (p3)	64 × 128	22	1	10	-	-	-	-	-	-	X
4	CCSM4	192 × 288	17	10	5	X	X	X	X	-	-	-
5	CNRM-CM5	128 × 256	17	10	5	X	X	X	-	X	-	-
6	CSIRO-Mk3.6.0 (p1)	96 × 192	18	5	6	X	X	X	-	X	-	-
7	CSIRO-Mk3.6.0 (p6)	96 × 192	18	5	6	-	-	-	-	-	X	-
8	FGOALS-g2	60 × 128	17	10	5	-	-	X	-	-	-	-
9	GFDL-CM3 (p1)	90 × 144	23	1	10	X	X	X	-	-	-	-
10	GFDL-CM3 (p2)	90 × 144	23	1	10	-	-	-	-	X	-	-
11	GFDL-ESM2-G	90 × 144	17	10	5	X	X	-	-	-	-	-
12	GFDL-ESM2-M (p1)	90 × 144	17	10	5	X	X	-	-	-	-	-
13	GFDL-ESM2-M (p2)	90 × 144	17	10	5	-	-	-	-	X	-	-
14	GFDL-ESM2-M (p7)	90 × 144	17	10	5	-	-	-	-	X	-	X
15	GFDL-ESM2-M (p8)	90 × 144	17	10	5	-	-	-	-	-	X	-
16	GISS-E2-H (p1)	90 × 144	17	10	5	-	-	X	-	-	-	-
17	GISS-E2-H (p109)	90 × 144	17	10	5	-	-	-	-	X	-	-
18	GISS-E2-R (p1)	90 × 144	17	10	5	X	X	X	-	-	-	-
19	GISS-E2-R (p2)	90 × 144	17	10	5	X	X	-	-	-	-	-
20	GISS-E2-R (p3)	90 × 144	17	10	5	-	-	X	-	-	-	-

Table S2 (continued): Information on horizonal resolution (latitude/longitude), the number of atmospheric levels (N_{H1})[1], the top model level (in hPa), the number of atmospheric levels above 100 hPa (N_{H2})[2], and the availability of simulation output for the CMIP-5 models used in this study (A = ALL+8.5; B = CTL; C = NAT; D = P1000; E = ANT; F = VOL; G = SOL).

	Model	Resolution	N_{H1}	Top	N_{H2}	A	B	C	D	E	F	G
21	GISS-E2-R (p102)	90 × 144	17	10	5	-	-	-	-	-	-	X
22	GISS-E2-R (p109)	90 × 144	17	10	5	-	-	-	-	X	-	-
23	GISS-E2-R (p121)	90 × 144	17	10	5	-	-	-	X	-	-	-
24	GISS-E2-R (p309)	90 × 144	17	10	5	-	-	-	-	X	-	-
25	HadGEM2-CC	144 × 192	23	0.4	11	X	X	-	-	-	-	-
26	HadGEM2-ES	144 × 192	17	10	5	X	X	X	-	-	-	-
27	INM-CM4	120 × 180	17	10	5	X	X	-	-	-	-	-
28	IPSL-CM5A-LR (p1)	96 × 96	17	10	5	X	X	X	X	-	-	-
29	IPSL-CM5A-LR (p2)	96 × 96	17	10	5	-	-	-	-	X	-	-
30	MIROC5	128 × 256	17	10	5	X	X	-	-	-	-	-
31	MIROC-ESM-CHEM	64 × 128	35	0.03	20	X	X	X	-	-	-	-
32	MIROC-ESM	64 × 128	35	0.03	20	X	X	X	X	-	-	-
33	MPI-ESM-LR	96 × 192	25	0.1	13	X	X	-	-	-	-	-
34	MPI-ESM-P	96 × 192	25	0.1	13	-	-	-	X	-	-	-
35	MRI-CGCM3	160 × 320	23	0.4	11	X	X	X	-	-	-	-
36	NorESM1-M	96 × 144	17	10	5	X	X	X	-	-	-	-
	Model total					20	20	16	6	8	2	3

[1]This represents the total number of model levels at which atmospheric temperature information is archived. It is typically smaller than the vertical resolution of the atmospheric model itself. Similarly, the top level in the fourth column is the highest pressure level at which atmospheric temperature data is archived – it is not necessarily identical to the pressure of the model lid.

[2]As in the case of N_{LEV1}, this is the number of atmospheric levels – above and not including the 100 hPa level – at which atmospheric temperature information is archived.

Table S3: Information on the external forcings which were included in the historical simulations of the CMIP-5 models used in this study. Information was extracted from the global attribute named "forcing" in the metadata of the relevant NetCDF files.[1] The third column uses the classification of *Eyring et al.* (2013) to stratify models according to their treatment of ozone (S8).

	Model	Forcing information from metadata	Ozone
1	BCC-CSM1.1	Nat, Ant, GHG, SD, Oz, Sl, Vl, SS, Ds, BC, OC	NOCHEM[2]
2	CanESM2	GHG, Oz, SA, BC, OC, LU, Sl, Vl	NOCHEM
3	CCSM4	Sl, GHG, Vl, SS, Ds, SD, BC, MD, OC, Oz, AA, LU	CHEM, S-OFF[3]
4	CNRM-CM5	GHG, SA, Sl, Vl, BC, OC[4]	CHEM, INT[5]
5	CSIRO-Mk3.6.0	Ant, Nat	NOCHEM
6	GFDL-CM3	GHG, SA, Oz, LU, Sl, Vl, SS, BC, MD, OC[6]	CHEM, INT
7	GFDL-ESM2G	GHG, SD, Oz, LU, Sl, Vl, SS, BC, MD, OC[7]	NOCHEM
8	GFDL-ESM2M	GHG, SD, Oz, LU, Sl, Vl, SS, BC, MD, OC[7]	NOCHEM
9	GISS-E2-R (p1)	GHG, LU, Sl, Vl, BC, OC, SA, Oz[8]	NOCHEM
10	GISS-E2-R (p2)	GHG, LU, Sl, Vl, BC, OC, SA, Oz[8]	CHEM, INT
11	HadGEM2-CC	GHG, Oz, SA, LU, Sl, Vl, BC, OC	NOCHEM
12	HadGEM2-ES	GHG, SA, Oz, LU, Sl, Vl, BC, OC[9]	NOCHEM
13	INM-CM4	GHG, Oz, SI, SA, Vl	NOCHEM
14	IPSL-CM5A-LR	Nat, Ant, GHG, SA, Oz, LU, SS, Ds, BC, MD, OC, AA	CHEM, S-OFF
15	MIROC5	GHG, SA, Oz, LU, Sl, Vl, SS, Ds, BC, MD, OC[10]	NOCHEM
16	MIROC-ESM-CHEM	GHG, SA, Oz, LU, Sl, Vl, MD, BC, OC	CHEM, INT
17	MIROC-ESM	GHG, SA, Oz, LU, Sl, Vl, MD, BC, OC	NOCHEM
18	MPI-ESM-LR	GHG, Oz, SD, Sl, Vl, LU	NOCHEM
19	MRI-CGCM3	GHG, SA, Oz, LU, Sl, Vl, BC, OC[11]	NOCHEM
20	NorESM1-M	GHG, SA, Oz, Sl, Vl, BC, OC	CHEM, S-OFF

Table S3 (continued): Information on the external forcings which were included in the historical simulations of the CMIP-5 models used in this study.

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[1]Forcing abbreviations are described in Appendix 1.2 of the CMIP-5 Data Reference Syntax document. Nat = natural forcing (a combination, not explicitly defined); Ant = anthropogenic forcing (a combination, not explicitly defined); GHG = well-mixed greenhouse gases; SD = anthropogenic sulfate aerosol (direct effects only); SI = anthropogenic sulfate aerosol (indirect effects only); SA = anthropogenic sulfate aerosol direct and indirect effects; Oz = tropospheric and stratospheric ozone; LU = land-use change; Sl = solar irradiance; Vl = volcanic aerosol; SS = sea salt; Ds = dust; BC = black carbon; MD = mineral dust; OC = organic carbon; AA = anthropogenic aerosols (a mixture of aerosols, not explicitly defined).

[2]NOCHEM = Model with prescribed changes in ozone.

[3]CHEM, S-OFF = Model with semi-offline ozone chemistry.

[4]Although stratospheric ozone forcing is not listed in the forcing metadata, chlorine concentration is an input for the prognostic ozone scheme of the `CNRM-CM5` model. So according to the CMIP-5 Data Reference Syntax document, stratospheric ozone should have been listed as an external forcing.

[5]CHEM, INT = Model with interactive ozone chemistry.

[6]GHG includes CO_2, CH_4, N_2O, CFC11, CFC12, HCFC22, and CFC113. Aerosol direct and indirect effects are included.

[7]GHG includes CO_2, CH_4, N_2O, CFC11, CFC12, HCFC22, and CFC113. "The direct effect of tropospheric aerosols is calculated by the model, but not the indirect effects."

[8]Also includes orbital change, BC on snow, and nitrate aerosols.

[9]GHG = CO_2, N_2O, CH_4, CFCs.

[10]GHG includes CO_2, N_2O, CH_4, and CFCs; Oz includes OH and H_2O_2; LU excludes change in lake fraction.

[11]GHG includes CO_2, CH_4, N_2O, CFC-11, CFC-12, and HCFC-22.

Table S4: Basic information relating to the start dates, end dates, and lengths (N_m, in months) of the CMIP-5 historical and RCP8.5 simulations used in this study. EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1]

	Model	EM	Hist. Start	Hist. End	Hist. N_m	RCP8.5 Start	RCP8.5 End	RCP8.5 N_m
1	BCC-CSM1.1	r1i1p1	1850-01	2012-12	1956	2006-01	2300-12	3540
2	CanESM2	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
3	CanESM2	r2i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
4	CanESM2	r3i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
5	CanESM2	r4i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
6	CanESM2	r5i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
7	CCSM4	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
8	CCSM4	r2i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
9	CCSM4	r3i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
10	CNRM-CM5	r1i1p1	1850-01	2005-12	1872	2006-01	2300-12	3540
11	CNRM-CM5	r2i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
12	CNRM-CM5	r4i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
13	CSIRO-Mk3.6.0	r10i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
14	GFDL-CM3	r1i1p1	1860-01	2005-12	1752	2006-01	2100-12	1140
15	GFDL-ESM2G	r1i1p1	1861-01	2005-12	1740	2006-01	2100-12	1140
16	GFDL-ESM2M	r1i1p1	1861-01	2005-12	1740	2006-01	2100-12	1140
17	GISS-E2-R (p1)	r1i1p1	1850-01	2005-12	1872	2006-01	2300-12	3540
18	GISS-E2-R (p2)	r1i1p2	1850-01	2005-12	1872	2006-01	2300-12	3540
19	HadGEM2-CC	r1i1p1	1859-12	2005-11	1752	2005-12	2099-12	1129
20	HadGEM2-CC	r2i1p1	1959-12	2005-12	553	2005-12	2099-12	1129
21	HadGEM2-CC	r3i1p1	1959-12	2005-12	553	2005-12	2099-12	1129
22	HadGEM2-ES	r1i1p1	1859-12	2005-11	1752	2005-12	2299-12	3529
23	HadGEM2-ES	r2i1p1	1859-12	2005-12	1753	2005-12	2100-11	1140
24	INM-CM4	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
25	IPSL-CM5A-LR	r2i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140

Table S4 (continued): Basic information relating to the start dates, end dates, and lengths of the CMIP-5 historical and RCP8.5 simulations used in this study.

	Model	EM	Hist. Start	Hist. End	Hist. (months)	RCP8.5 Start	RCP8.5 End	RCP8.5 (months)
26	MIROC5	r1i1p1	1850-01	2012-12	1956	2006-01	2100-12	1140
27	MIROC-ESM-CHEM	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
28	MIROC-ESM	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
29	MPI-ESM-LR	r1i1p1	1850-01	2005-12	1872	2006-01	2300-12	3540
30	MPI-ESM-LR	r2i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
31	MPI-ESM-LR	r3i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
32	MRI-CGCM3	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140
33	NorESM1-M	r1i1p1	1850-01	2005-12	1872	2006-01	2100-12	1140

[1]See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

Table S5: Basic information regarding CMIP-5 simulations with estimated historical changes in natural external forcing (NAT). The simulations listed in the Table were used for estimating V_{TOT}, the 'total' natural variability. EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1] The total number of simulated months (N_m) is given in column 5.

	Model	EM	Start	End	N_m	Forcing[2]
1	BCC-CSM1.1	r1i1p1	1850-01	2012-12	1956	Sl, Vl
2	CanESM2	r1i1p1	1850-01	2012-12	1956	Sl, Vl
3	CCSM4	r1i1p1	1850-01	2005-12	1872	Sl, Vl[3]
4	CNRM-CM5	r1i1p1	1850-01	2012-12	1956	Sl, Vl
5	CSIRO-Mk3.6.0	r1i1p1	1850-01	2012-12	1956	Nat
6	FGOALS-g2	r1i1p1	1850-01	2009-12	1920	Sl
7	GFDL-CM3	r1i1p1	1860-01	2005-12	1752	Vl, Sl[4]
8	GISS-E2-H (p1)	r1i1p1	1850-01	2012-12	1956	Sl, Vl[5]
9	GISS-E2-R (p1)	r1i1p1	1850-01	2012-12	1956	Sl, Vl[5]
10	GISS-E2-R (p3)	r1i1p3	1850-01	2012-12	1956	Sl, Vl[5]
11	HadGEM2-ES	r1i1p1	1860-01	2018-12	1908	Sl, Vl
12	IPSL-CM5A-LR	r1i1p1	1850-01	2005-12	1872	Nat[6]
13	MIROC-ESM	r1i1p1	1850-01	2005-12	1872	Sl, Vl
14	MIROC-ESM-CHEM	r1i1p1	1850-01	2005-12	1872	Sl, Vl
15	MRI-CGCM3	r1i1p1	1850-01	2005-12	1872	Sl, Vl
16	NorESM1-M	r1i1p1	1850-01	2012-12	1956	Sl, Vl

Table S5 (continued): Basic information regarding CMIP-5 simulations with estimated historical changes in natural external forcing (NAT).

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[1]See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

[2]Forcing information was extracted directly from the global attribute named “forcing” in the metadata of NetCDF files. Forcing abbreviations are described in Appendix 1.2 of the CMIP-5 Data Reference Syntax document. Nat = natural forcing (a combination, not explicitly defined); Ant = anthropogenic forcing (a combination, not explicitly defined); GHG = well-mixed greenhouse gases; SD = anthropogenic sulfate aerosol (direct effects only); SI = anthropogenic sulfate aerosol (indirect effects only); SA = anthropogenic sulfate aerosol direct and indirect effects; Oz = tropospheric and stratospheric ozone; LU = land-use change; Sl = solar irradiance; Vl = volcanic aerosol; SS = sea salt; Ds = dust; BC = black carbon; MD = mineral dust; OC = organic carbon; AA = anthropogenic aerosols (a mixture of aerosols, not explicitly defined).

[3]Note that SS, Ds, SD, BC, MD, OC, Oz, AA, LU were ”all fixed at or cycled over 1850 values.”

[4]Includes dynamic vegetation (1860 land use).

[5]Also includes orbital change.

[6]In the metadata for the `IPSL-CM5A-LR` historicalNat simulation, the forcing is mistakenly listed as: “Nat, Ant, GHG, SA, Oz, LU, SS, Ds, BC, MD, OC, AA.”

Table S6: Basic information regarding CMIP-5 "Last Millennium" (P1000) simulations. In most cases, these simulations involve natural external forcing, generally over the period from 850 to 1849 or 1850. The simulations listed in the Table were used for estimating V_{TOT}, the 'total' natural variability. EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1] The total number of simulated months (N_m) is given in column 5.

	Model	EM	Start	End	N_m	Forcing[2]
1	BCC-CSM1.1	r1i1p1	850-01	2000-12	13812	Nat, Ant[3]
2	CCSM4	r1i1p1	850-01	1850-12	12012	Sl, GHG, Vl, LU[4]
3	GISS-E2-R	r1i1p121	850-01	1850-12	12012	GHG, LU, Sl, Vl[5]
4	IPSL-CM5A-LR	r1i1p1	850-01	1850-12	12012	Nat, Ant, GHG
5	MIROC-ESM	r1i1p1	850-01	1849-12	12000	GHG, LU, Sl, Vl[6]
6	MPI-ESM-P	r1i1p1	850-01	1849-12	12000	GHG, Sl, Vl, LU

[1] See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

[2] Forcing information was extracted directly from global attribute named "forcing" in the metadata of NetCDF files. Forcing abbreviations are described in Appendix 1.2 of the CMIP-5 Data Reference Syntax document. Nat = natural forcing (a combination, not explicitly defined); Ant = anthropogenic forcing (a combination, not explicitly defined); GHG = well-mixed greenhouse gases; SD = anthropogenic sulfate aerosol (direct effects only); SI = anthropogenic sulfate aerosol (indirect effects only); SA = anthropogenic sulfate aerosol direct and indirect effects; Oz = tropospheric and stratospheric ozone; LU = land-use change; Sl = solar irradiance; Vl = volcanic aerosol; SS = sea salt; Ds = dust; BC = black carbon; MD = mineral dust; OC = organic carbon; AA = anthropogenic aerosols (a mixture of aerosols, not explicitly defined).

[3] Nat, Ant: "includes Vl, Sl, CO_2, CH_4, N_2O, orbital parameters, and (since 1850) forcing by anthropogenic aerosols."

[4] All four forcings are "time-varying over course of simulation". Note that SS, Ds, SD, BC, MD, OC, Oz, AA, LU were "all fixed at or cycled over 1850 values."

[5] Also includes orbital forcing.

[6] Also includes orbital forcing. Note that: "GHG includes only CH_4 and N_2O; CO_2 is predicted by model."

Table S7: Basic information regarding CMIP-5 simulations with estimated historical changes in anthropogenic forcing only. The simulations listed in the Table were used for estimating the ANT fingerprint. EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1] The total number of simulated months (N_m) is given in column 5.

	Model	EM	Start	End	N_m	Forcing
1	CNRM-CM5	r1i1p1	1850-01	2012-12	1956	GHG, SA, BC, OC[2]
2	CSIRO-Mk3.6.0	r1i1p1	1850-01	2012-12	1956	Ant
3	GFDL-CM3	r1i1p2	1860-01	2005-12	1752	GHG, SA, Oz, LU, SS, BC MD, OC[3]
4	GFDL-ESM2M	r1i1p2	1861-01	2005-12	1740	GHG, SD, Oz, LU, SS, BC MD, OC[4]
5	GISS-E2-H (p109)	r1i1p109	1850-01	2012-12	1956	GHG, LU, BC, OC, SA, Oz[5]
6	GISS-E2-R (p109)	r1i1p109	1850-01	2012-12	1956	GHG, LU, BC, OC, SA, Oz[5]
7	GISS-E2-R (p309)	r1i1p309	1850-01	2012-12	1956	GHG, LU, BC, OC, SA, Oz[5]
8	IPSL-CM5A-LR	r1i1p2	1850-01	2005-12	1872	Ant

[1] See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

[2] Forcing information was extracted directly from the global attribute named "forcing" in the metadata of NetCDF files. Forcing abbreviations are described in Appendix 1.2 of the CMIP-5 Data Reference Syntax document. Nat = natural forcing (a combination, not explicitly defined); Ant = anthropogenic forcing (a combination, not explicitly defined); GHG = well-mixed greenhouse gases; SD = anthropogenic sulfate aerosol (direct effects only); SI = anthropogenic sulfate aerosol (indirect effects only); SA = anthropogenic sulfate aerosol direct and indirect effects; Oz = tropospheric and stratospheric ozone; LU = land-use change; Sl = solar irradiance; Vl = volcanic aerosol; SS = sea salt; Ds = dust; BC = black carbon; MD = mineral dust; OC = organic carbon; AA = anthropogenic aerosols (a mixture of aerosols, not explicitly defined).

[3] GHG includes CO_2, CH_4, N_2O, CFC11, CFC12, HCFC22, and CFC113. Aerosol direct and indirect effects are included.

[4] GHG includes CO_2, CH_4, N_2O, CFC11, CFC12, HCFC22, and CFC113. "The direct effect of tropospheric aerosols is calculated by the model, but not the indirect effects."

[5] Also includes BC on snow, nitrate aerosols.

Table S8: Basic information regarding CMIP-5 simulations with estimated historical changes in volcanic forcing only (VOL) and in solar forcing only (SOL). EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1] The total number of simulated months (N_m) is given in column 5.

	Model	EM	Type	Start	End	N_m	Forcing
1	CSIRO-Mk3.6.0 (p6)	r1i1p6	VOL	1850-01	2012-12	1956	Vl
2	GFDL-ESM2M (p8)	r1i1p8	VOL	1861-01	2005-12	1740	Vl
1	CanESM2 (p3)	r1i1p3	SOL	1850-01	2012-12	1956	Sl
2	GFDL-ESM2M (p7)	r1i1p7	SOL	1861-01	2005-12	1740	Sl
3	GISS-E2-R (p102)	r1i1p102	SOL	1850-01	2012-12	1956	Sl

[1] See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

[2] Forcing information was extracted directly from the global attribute named "forcing" in the metadata of NetCDF files. Forcing abbreviations are described in Appendix 1.2 of the CMIP-5 Data Reference Syntax document. Nat = natural forcing (a combination, not explicitly defined); Ant = anthropogenic forcing (a combination, not explicitly defined); GHG = well-mixed greenhouse gases; SD = anthropogenic sulfate aerosol (direct effects only); SI = anthropogenic sulfate aerosol (indirect effects only); SA = anthropogenic sulfate aerosol direct and indirect effects; Oz = tropospheric and stratospheric ozone; LU = land-use change; Sl = solar irradiance; Vl = volcanic aerosol; SS = sea salt; Ds = dust; BC = black carbon; MD = mineral dust; OC = organic carbon; AA = anthropogenic aerosols (a mixture of aerosols, not explicitly defined).

Table S9: Basic information relating to the start dates, end dates, and lengths (N_m, in months) of the CMIP-5 pre-industrial control runs used in this study. EM is the "ensemble member" identifier described in the CMIP-5 Data Reference Syntax document.[1]

	Model	EM	Start	End	N_m
1	BCC-CSM1.1	r1i1p1	1800-01	2799-12	6000
2	CanESM2	r1i1p1	2015-01	3010-12	11952
3	CCSM4	r1i1p1	800-01	1300-12	6012
4	CNRM-CM5	r1i1p1	1850-01	2699-12	10200
5	CSIRO-Mk3.6.0	r1i1p1	1651-01	2150-12	6000
6	GFDL-CM3	r1i1p1	1-01	500-12	6000
7	GFDL-ESM2G	r1i1p1	1-01	500-12	6000
8	GFDL-ESM2M	r1i1p1	1-01	500-12	6000
9	GISS-E2-R (p1)	r1i1p1	3981-01	4530-12	6600
10	GISS-E2-R (p2)	r1i1p2	3590-01	4120-12	6372
11	HadGEM2-CC	r1i1p1	1859-12	2099-12	2881
12	HadGEM2-ES	r1i1p1	1859-12	2435-11	6912
13	INM-CM4	r1i1p1	1850-01	2349-12	6000
14	IPSL-CM5A-LR	r1i1p1	1800-01	2799-12	12000
15	MIROC5	r1i1p1	2000-01	2669-12	8040
16	MIROC-ESM-CHEM	r1i1p1	1846-01	2100-12	3060
17	MIROC-ESM	r1i1p1	1800-01	2330-12	6372
18	MPI-ESM-LR	r1i1p1	1850-01	2849-12	12000
19	MRI-CGCM3	r1i1p1	1851-01	2350-12	6000
20	NorESM1-M	r1i1p1	700-01	1200-12	6012

[1] See http://cmip-pcmdi.llnl.gov/cmip5/documents.html for further details.

Figure S1: Time series of simulated monthly-mean, near-global anomalies in the temperature of the lower stratosphere (TLS). Results are from simulations with historical changes in solar and volcanic forcing (NAT) performed with 16 different CMIP-5 models (panels A-P). Anomalies were averaged over 82.5°N-82.5°S (the latitudinal extent of RSS TLS data), and are defined with respect to climatological monthly means over 1861 to 1870. The y-axis range is identical in each panel. Note that the `FGOALS-G2` and `IPSL-CM5A-LR` models lack volcanically-induced lower stratospheric warming signals: the `FGOALS-G2` NAT simulation includes solar forcing only, while `IPSL-CM5A-LR` has only indirect representation of volcanic effects on surface and tropospheric temperature (through tuning of the solar irradiance).

Figure S2: As for Fig. S1, but for time series of monthly-mean, near-global anomalies in the temperature of the lower troposphere (TLT; panels A-P). Model anomalies are averaged over 82.5°N-70°S (the latitudinal extent of RSS TLT data).

Figure S3: Changes in the climatological seasonal cycle of lower stratospheric temperature (TLS) in the CMIP-5 ALL+8.5 (panel A) and NAT simulations (panel B). Results are for spatial averages over a domain encompassing the Antarctic ozone hole (60°S-82.5°S). Climatological monthly means were calculated using the ALL+8.5 and NAT multi-model averages, which were based on 20 and 16 models (respectively). The reference climatological monthly means are for the period 1861 to 1880. Temporal changes in the seasonal cycle (for non-overlapping 20-year periods) are expressed as departures from the reference climatological monthly means. For the ALL+8.5

case, the last analysis period is 2001 to 2012 (since 2012 is the end of the satellite record). The NAT runs end in 2005, so the last NAT analysis period is 2001 to 2005. Stratospheric ozone depletion is the main driver of the overall cooling trend in the ALL+8.5 climatological seasonal cycle, and of the progressive changes in its shape. There are no large temporal changes in the seasonal cycle in the multi-model average of the NAT simulations. Note that the y-axis range is identical in panels A and B.

Figure S4: Zonal-mean atmospheric temperature trends over January 1979 to December 2012 in ALL+8.5 simulations performed with 20 individual CMIP-5 models (panels A-T). Trends were calculated after first regridding model and observational TLS, TMT, and TLT anomaly data to a $5^\circ \times 5^\circ$ latitude/longitude grid, and then computing zonal averages. Anomalies were defined relative to climatological monthly means over the trend analysis period. Results are plotted in "MSU space", at the approximate peaks of the TLS, TMT, and TLT global-mean MSU weighting functions (74, 595, and 740 hPa, respectively). For models with multiple realizations of the ALL+8.5 experiment, only the first realization is shown.

Figure S5: Time series of monthly-mean, near-global anomalies in TLS, TMT, and TLT in the NAT simulations (panels A-C). Results are for the 324-month period from January 1979 to December 2005 – the period of overlap between the satellite atmospheric temperature record and the NAT simulations. The El Chichón and Pinatubo eruptions occurred 3 and 12 years after the start of the satellite record (respectively). The peak of the volcanically-induced tropospheric cooling signal caused by Pinatubo is close to the mid-point of the 324-month analysis period; the peak tropospheric cooling arising from El Chichón is near the beginning of this period. This asymmetry in the distribution of volcanic signals (relative to the mid-point of the 1979 to 2005 analysis period) leads to the small positive TLT and TMT trends in the NAT and VOL simulations, and to the small negative trend in TLS. Analysis of the NAT temperature changes over a longer period (1950 to 2005) yields the opposite result: slight cooling of the troposphere and warming of the lower stratosphere.

Figure S6: Time series of simulated monthly-mean, near-global anomalies in the temperature of the lower stratosphere (TLS; panels A-F) and the lower troposphere (TLT; panels G-L). Results are from "P1000" simulations with estimated changes in solar and volcanic forcing (S9) over the period 850 to 1849. TLS and TLT are spatially averaged over the same regions used in Figures S1 and S2 (respectively). Anomalies are defined with respect to climatological monthly means over 850 to 1700. To faciliate visual comparison of inter-model differences in the naturally forced TLS (TLT) signals, the y-axis range is identical in panels A-F (G-L). The slow secular drift in `MIROC-ESM` TLT anomalies is due to inadequate spin-up.

Figure S7: As for Figure 4 of the main text, but for signal trends (panel A), noise trends (panel B) and S/N ratios (panel C) calculated with the ALL+8.5 fingerprint (Fig. 3B) rather than the ANT fingerprint (Fig. 3A). The ALL+8.5 fingerprint was computed using the multi-model average zonal-mean changes in atmospheric temperature over 1861 to 2012. For all other processing details, refer to Fig. 4.

Figure S8: The effect of the number of layers of atmospheric temperature on D&A results. The "3-layer" results involve changes in zonally averaged TLS, TMT, and TLT. The signal trends for the 3-layer case (panel A) are identical to those shown in *SI Appendix* Fig. S7A. The noise trends (panel B) and S/N ratios (panel C) for the 3-layer analysis correspond to the "CTL noise" results given in *SI Appendix* Figs. S7B and C. In the 2-layer analysis, only zonal-mean changes in TLS and TLT were used. The 3- and 2-layer analyses use the same simulations for estimating the fingerprint (the ALL+8.5 runs). Both analyses also employ the same CTL simulations for noise estimation. Because S/N ratios are larger for stratospheric than for tropospheric temperature changes (S3), S/N ratios are enhanced by removal of the TMT results, and hence are larger in the 2-layer case.

Figure S9: S/N ratios calculated with the ALL+8.5 fingerprint (Fig. 3A) and CTL noise. Estimates of the ALL+8.5 fingerprint strength in observations are identical to the "model-versus-observed" S/N ratios given in *SI Appendix* Fig. S7C. "Model-versus-model" S/N ratios are also shown. In the latter cases, the ensemble-mean ALL+8.5 temperature changes over 1979 to 2012 for each of 20 different CMIP-5 models (and for the ALL+8.5 multi-model average) were projected onto the ALL+8.5 multi-model average fingerprint. The "model-versus-model" S/N ratios are generally larger than the "model-versus-observed" S/N results. This bias in S/N ratios is primarily due to the fact that most ALL+8.5 models overestimate the observed tropospheric warming during the satellite era (S3).

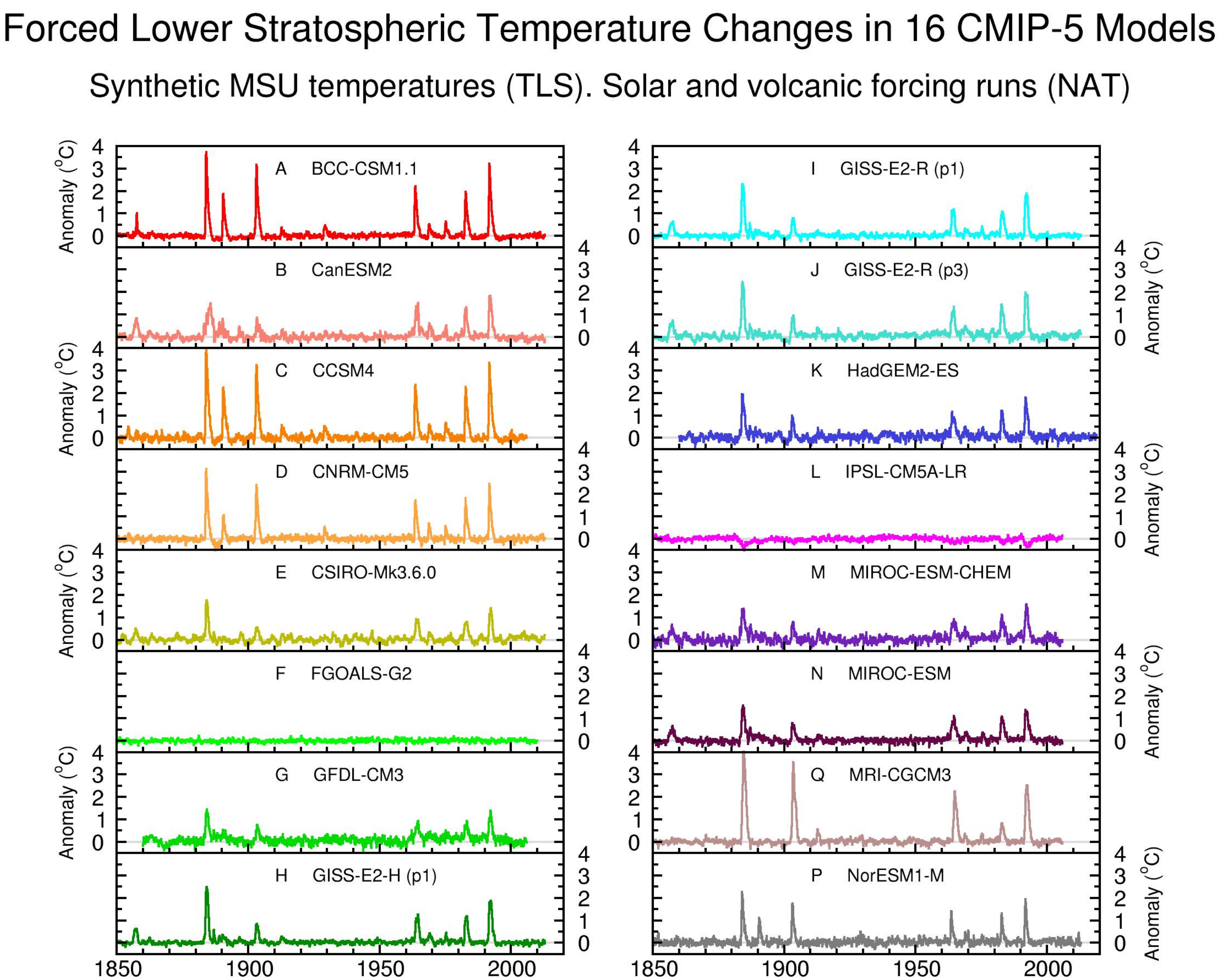

Figure S1: Santer *et al.*

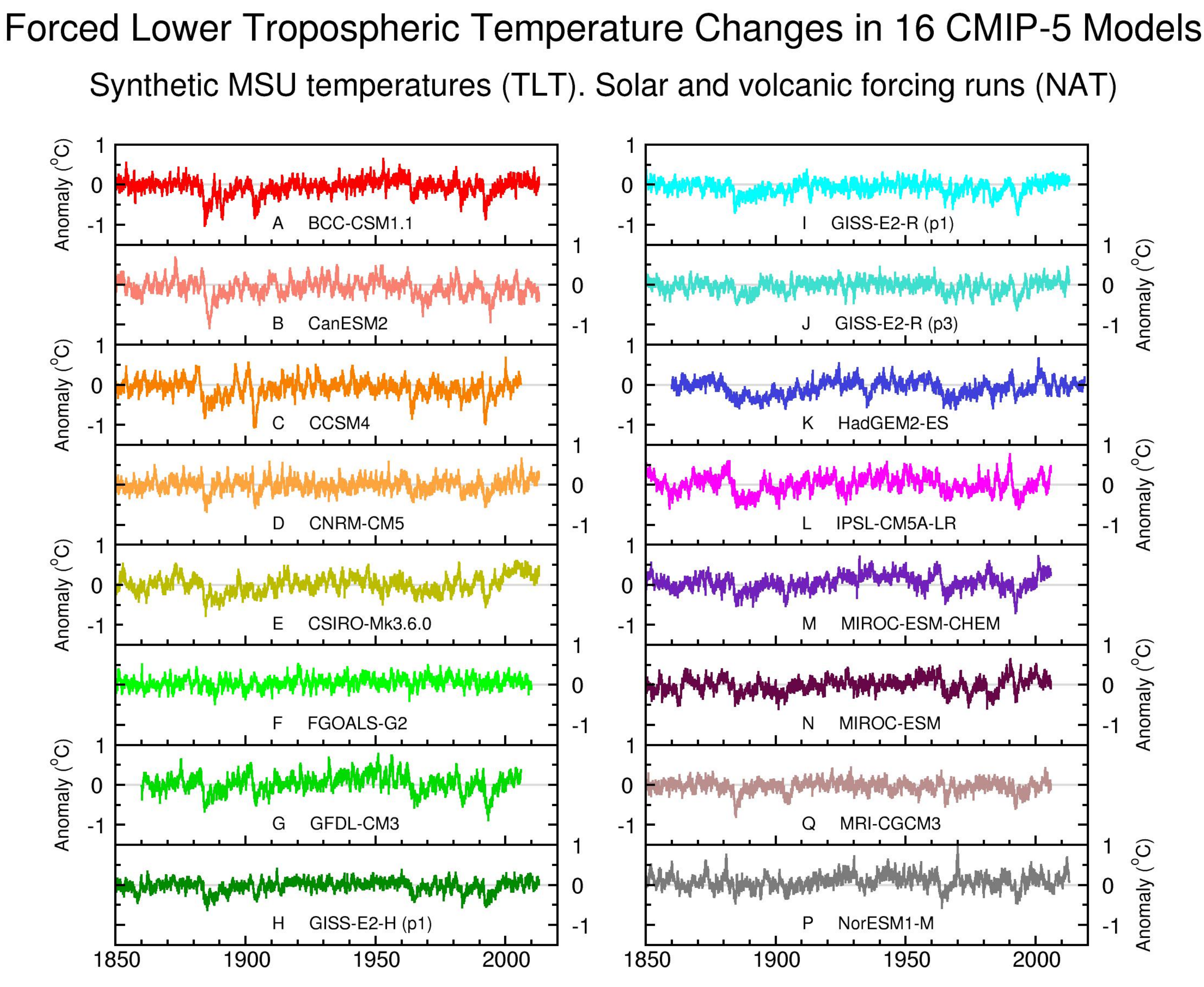

Figure S2: Santer *et al.*

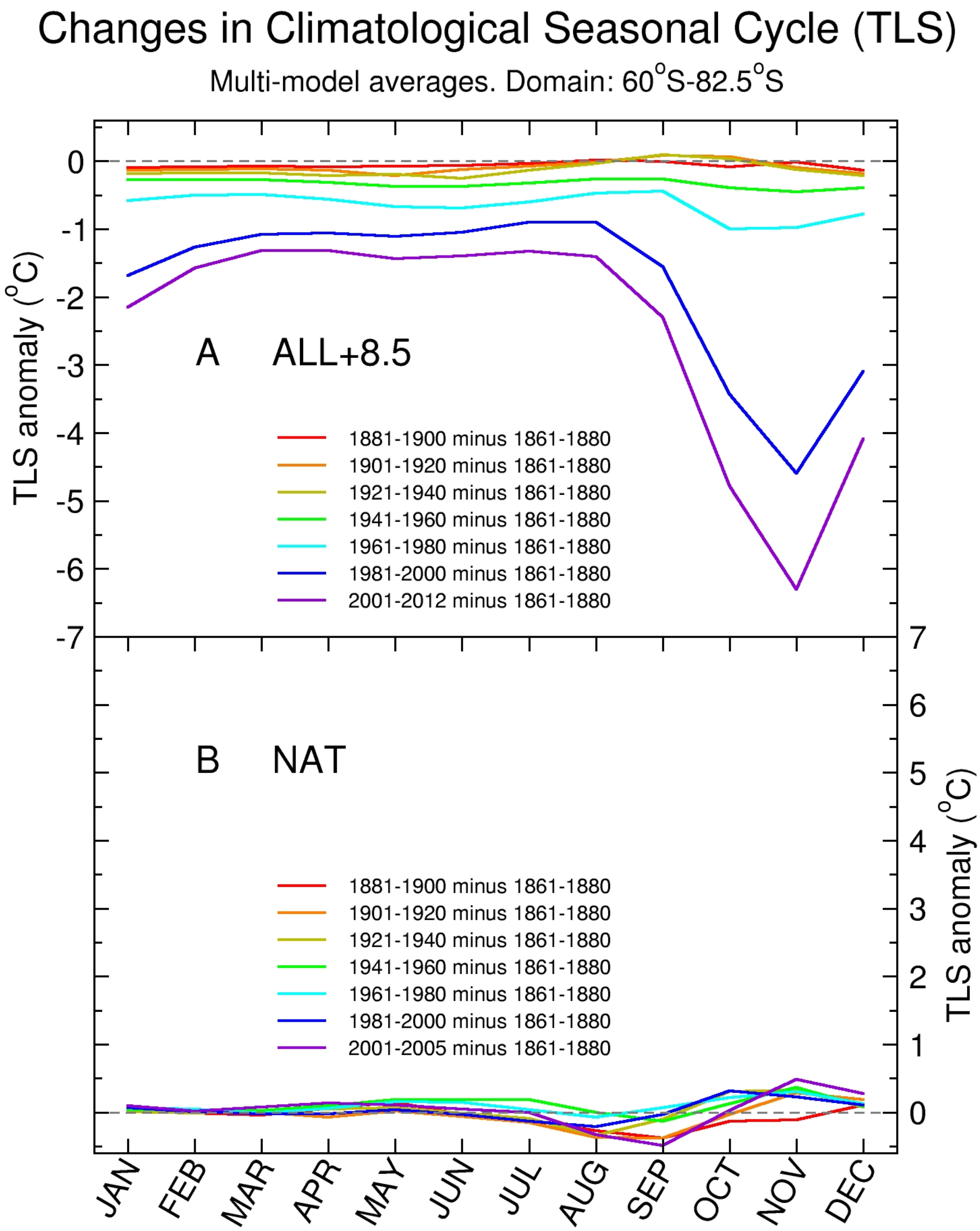

Figure S3: Santer *et al.*

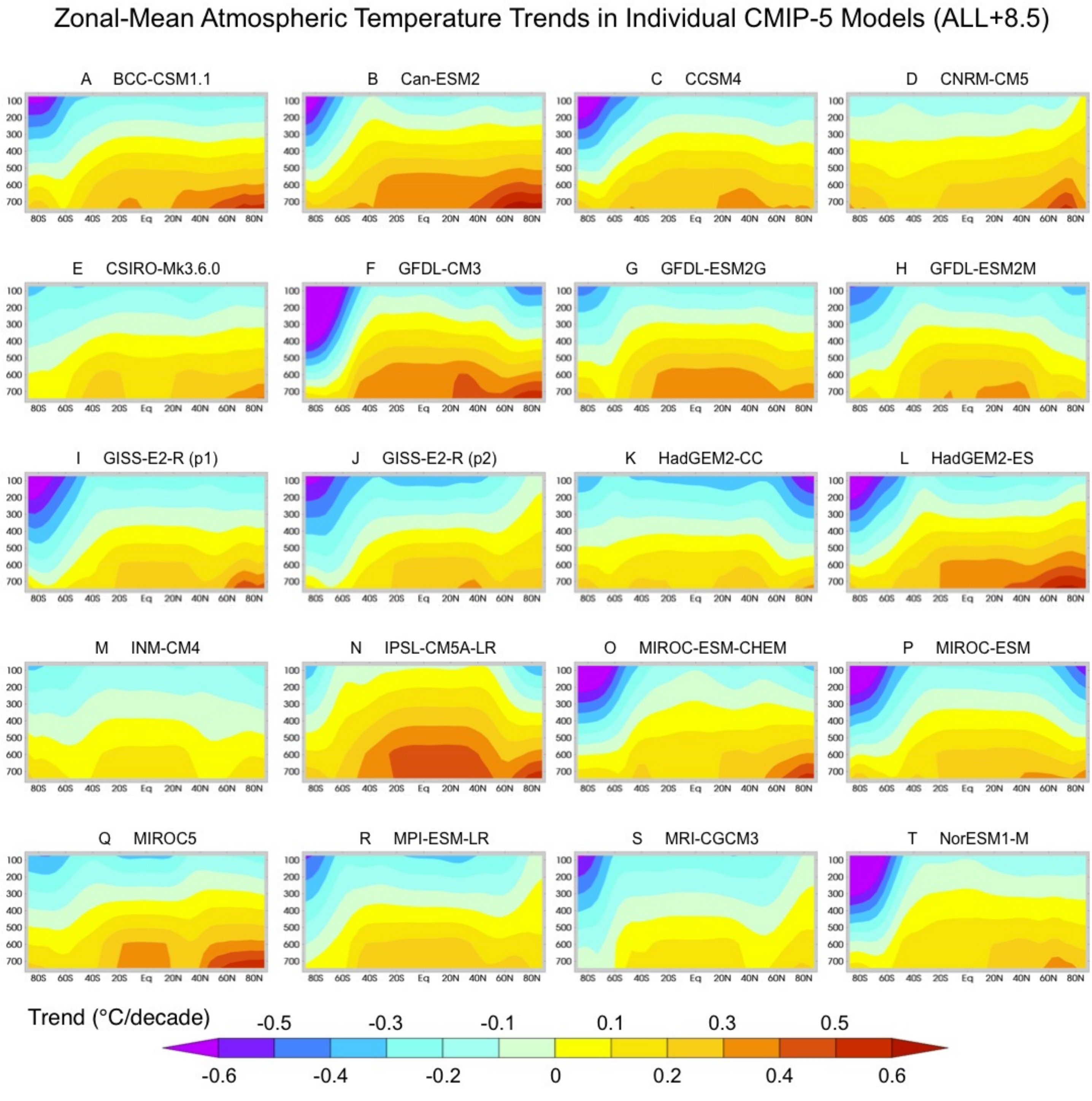

Figure S4: Santer *et al.*

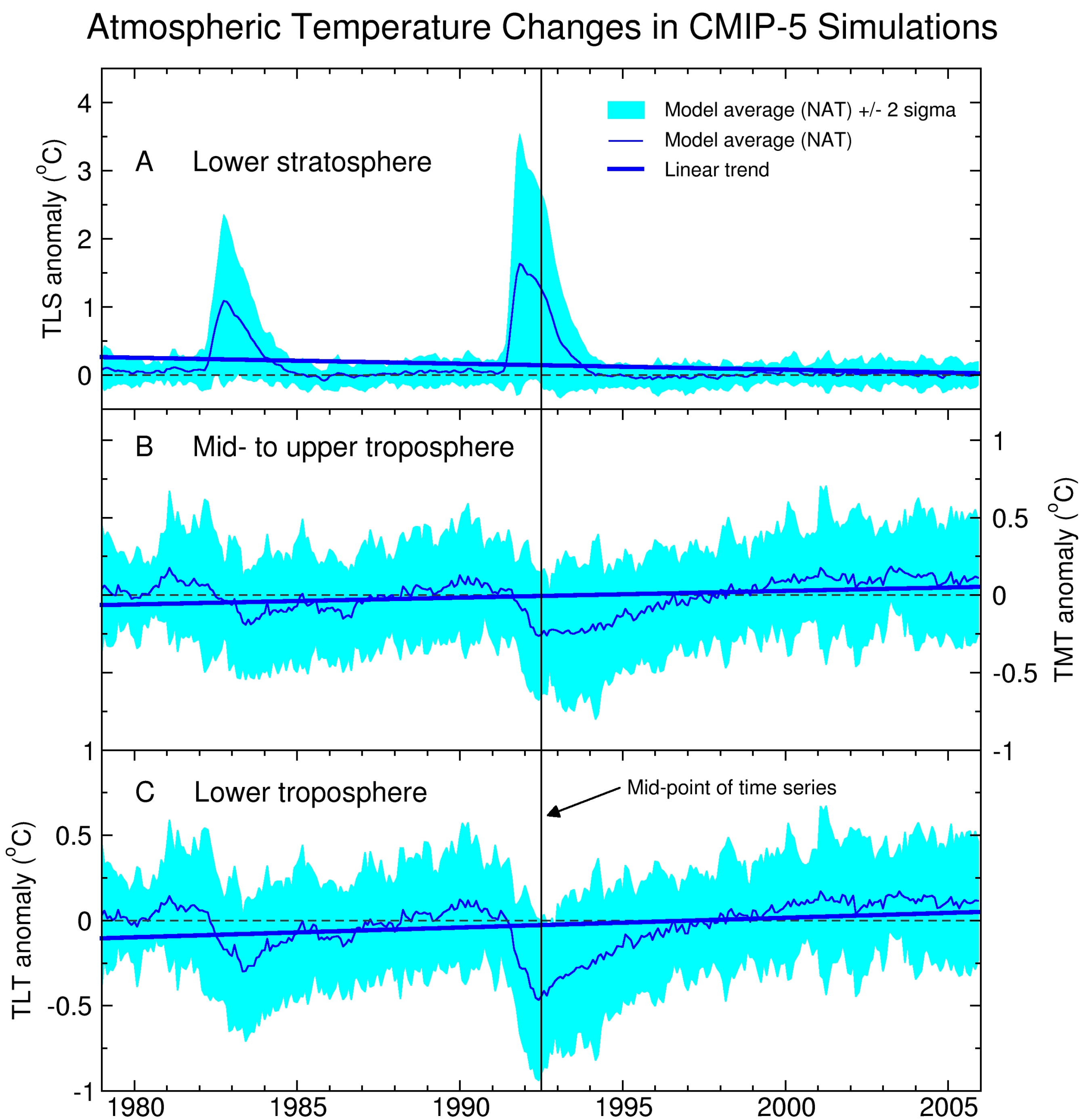

Figure S5: Santer *et al.*

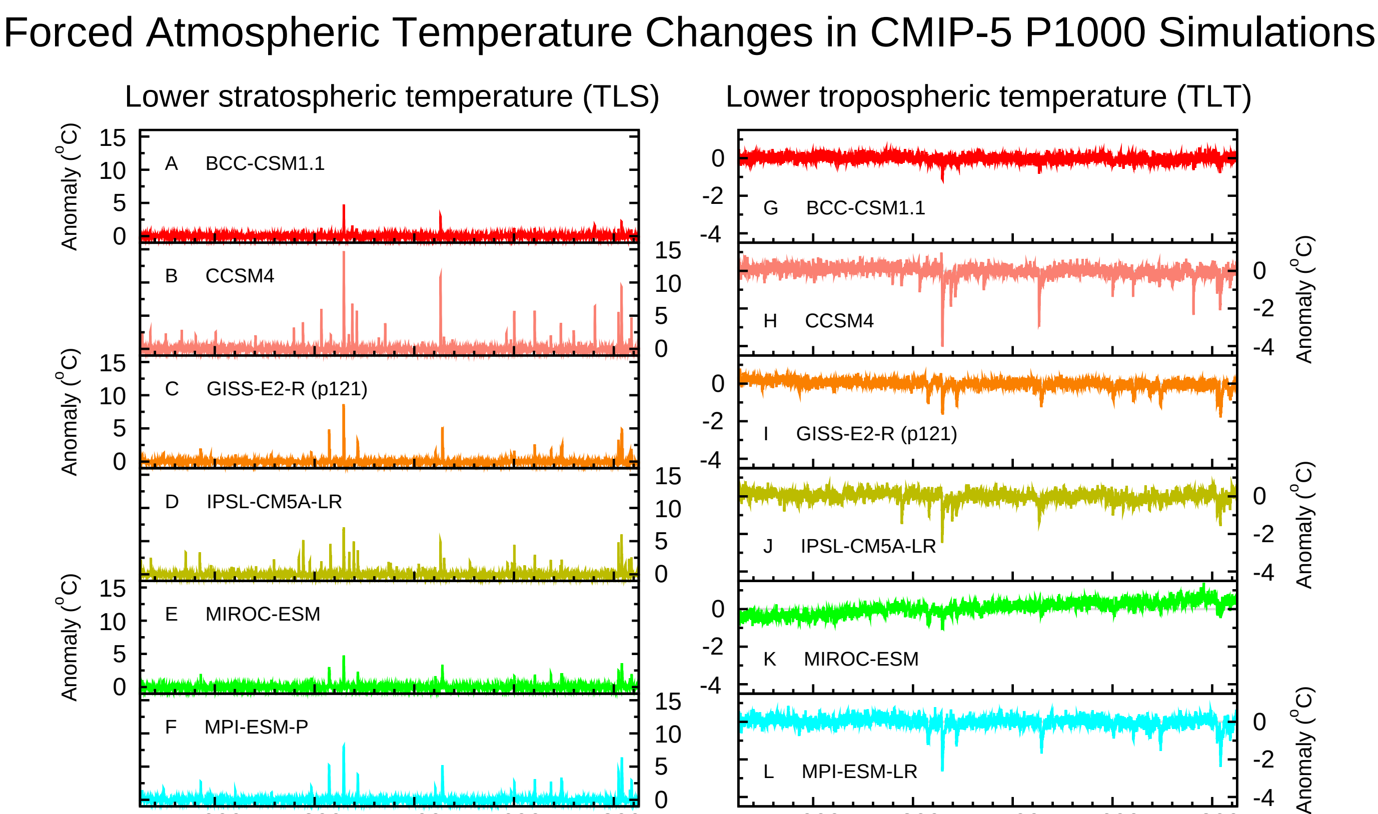

Figure S6: Santer *et al.*

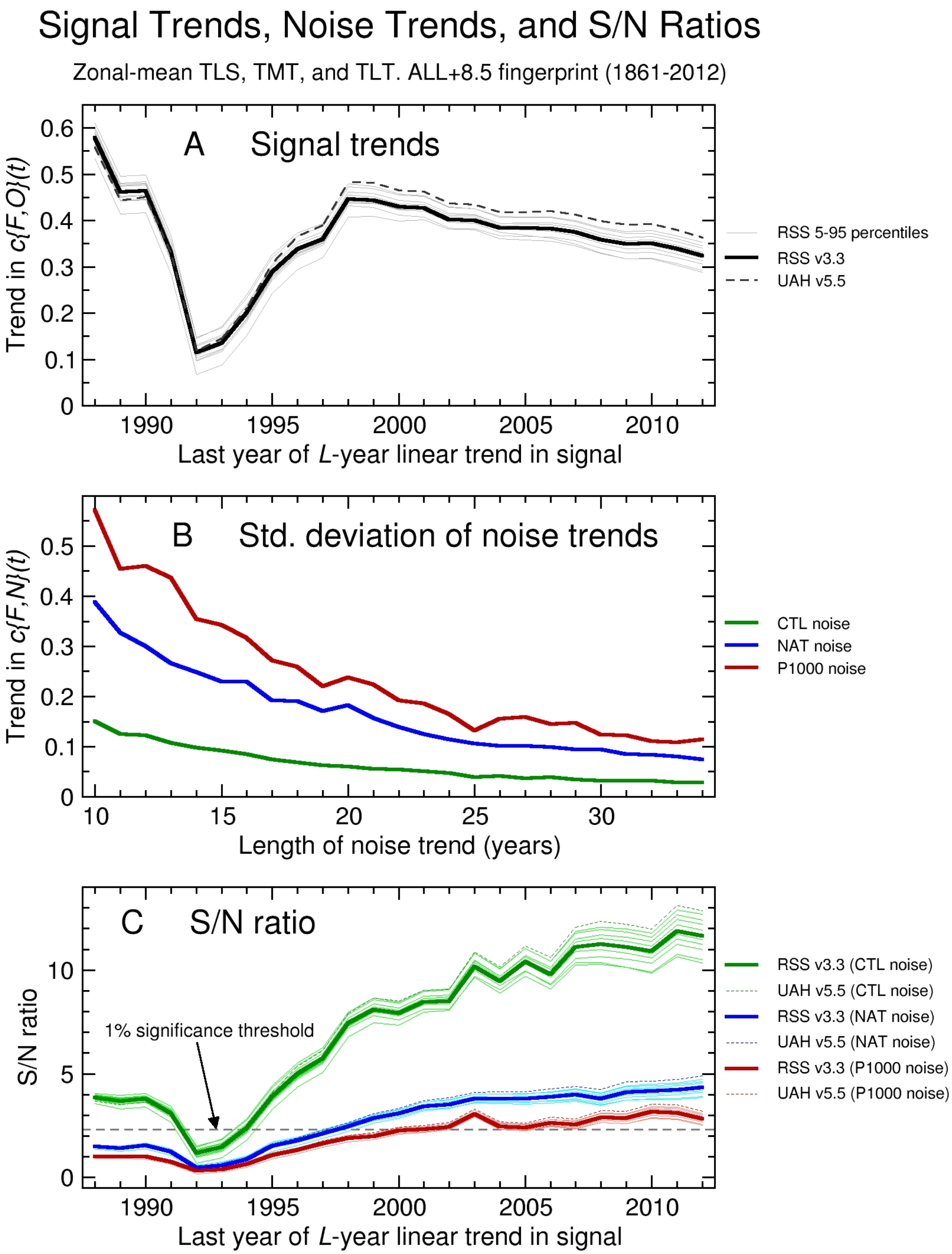

Figure S7: Santer *et al.*

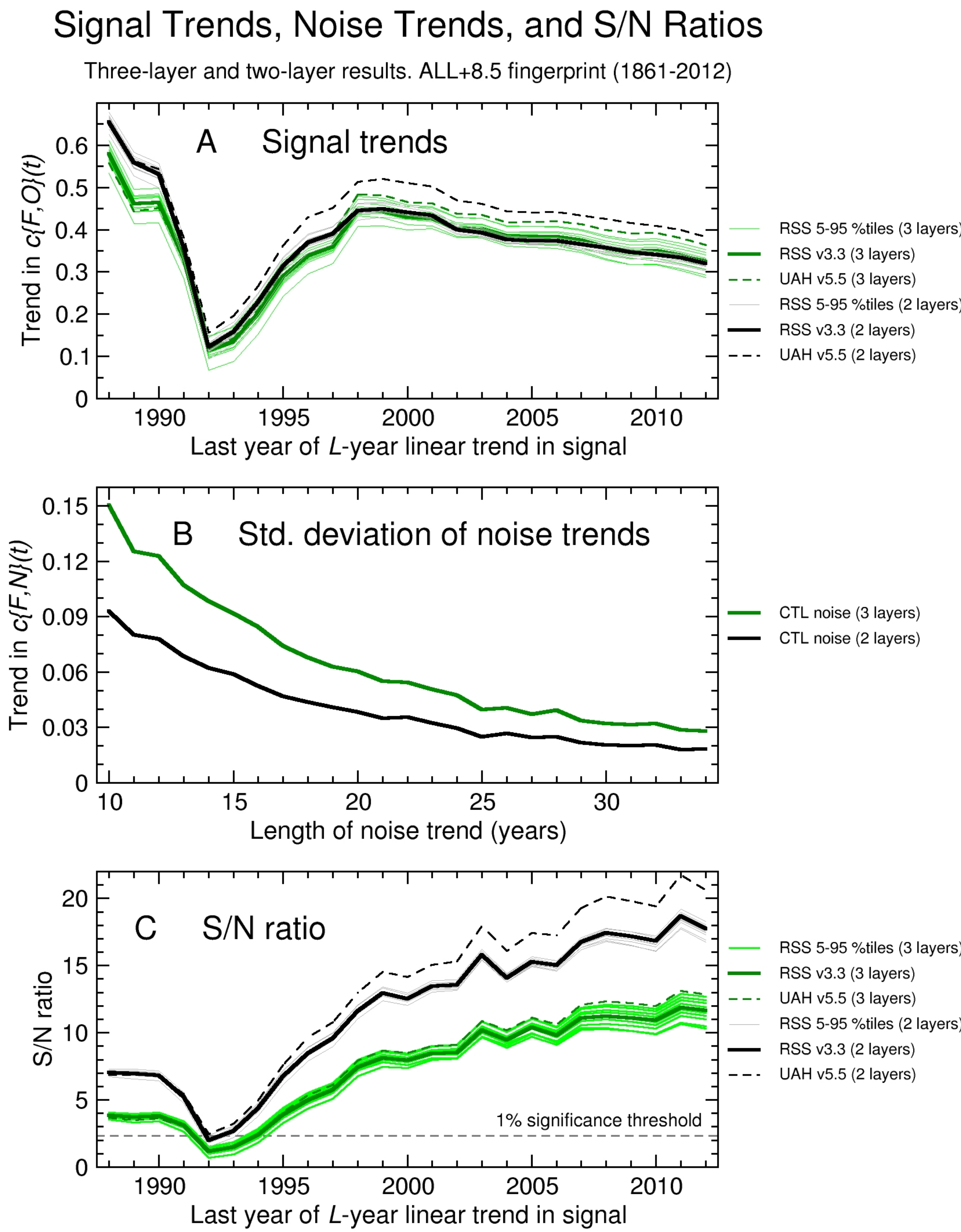

Figure S8: Santer *et al.*

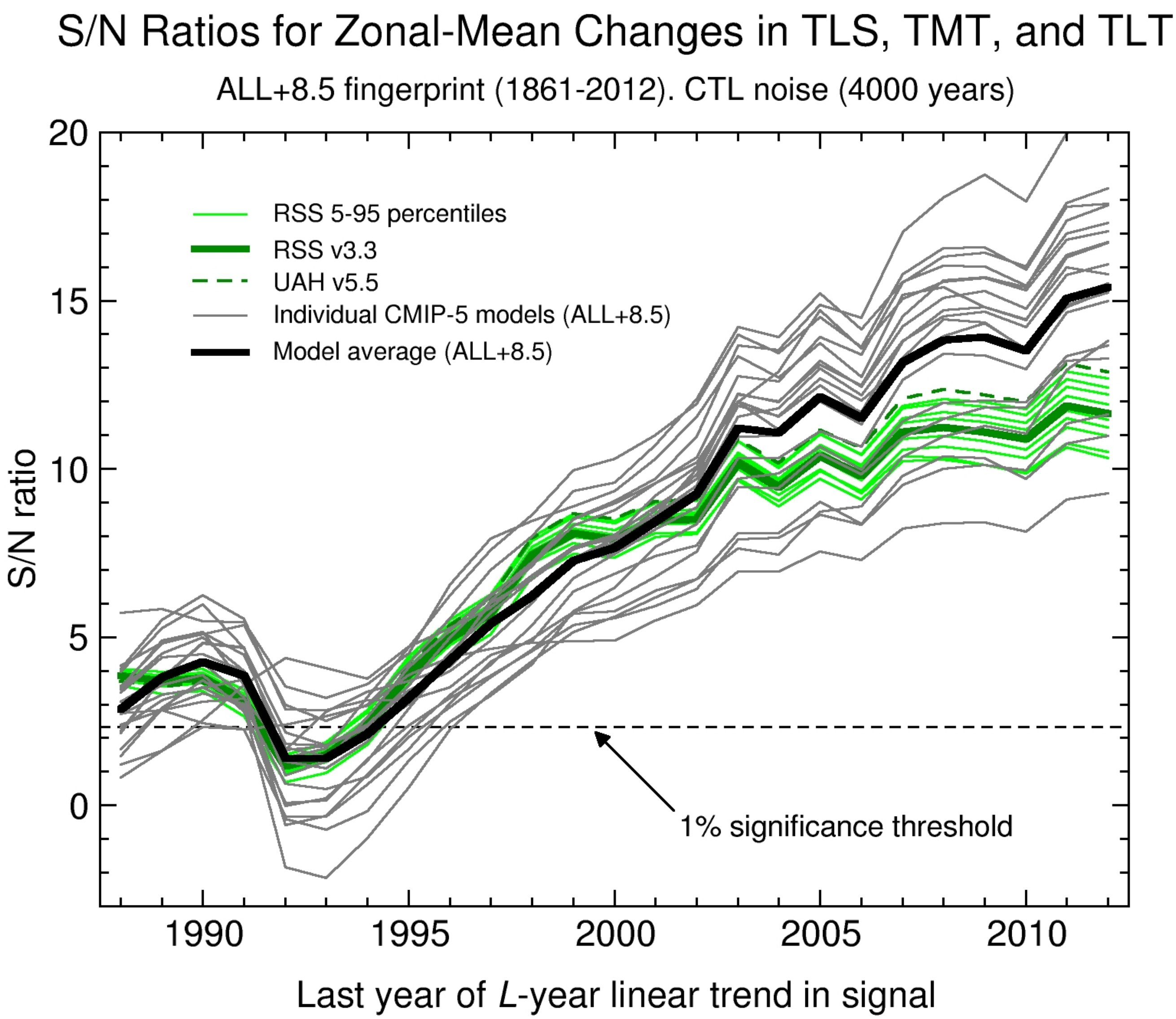

Figure S9: Santer *et al.*